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Shower/commode wheeled chair

Abstract

A shower/commode wheeled chair adapted to be locatable over a commode, the shower/commode wheeled chair comprising a frame, a seat support portion, a front portion extending downwardly from the seat support portion, a back support, a shower/commode seat, wheels and a footrest coupled to the front portion.

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Background/Summary

RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application No. 63/407,495, filed Sep. 16, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(1) In general, this disclosure is directed to a shower/commode wheeled chair for use by patients and elderly and/or injured individuals in conjunction with movement or positioning relative to a shower or commode/toilet.

BACKGROUND

(2) Shower/commode wheeled chairs have been used to help position hospital patients and elderly and/or injured individuals as needed to perform certain routine, daily activities, such as showering and or use of bathroom facilities. There is a need for improvement in the adjustability, storage, and/or flexibility in the use of such devices.

SUMMARY

(3) In some embodiments of this disclosure, a shower/commode wheeled chair is adapted to be locatable over a commode, the shower/commode wheeled chair comprising a frame, a seat support portion, a front portion extending downwardly from the seat support portion, a back support, a shower/commode seat, wheels and a footrest coupled to the front portion, wherein the footrest is configured to pivot about a generally horizontal axis. In some embodiments, the footrest may also be (or may alternately be) configured to rotate about a generally vertical axis.

(4) In some embodiments of this disclosure, a shower/commode wheeled chair is adapted to be locatable over a commode, the shower/commode wheeled chair comprising a frame, a seat support portion, a front portion extending downwardly from the seat support portion, a back support, a shower/commode seat, wheels and one or more arm rests having an attachment portion with a catch/latch mechanism, and an arm support portion, wherein the arm support portion includes a spring release mechanism that enables pivoting of the arm support portion relative to the attachment portion by displacing the arm support portion a horizontal distance.

(5) In some embodiments, of this disclosure, a shower/commode wheeled chair is adapted to be locatable over a commode, the shower/commode wheeled chair comprising a frame, a seat support portion, a front portion extending downwardly from the seat support portion, a back support, a shower/commode seat, and a plurality of wheels including at least two rear wheels, wherein at least one of the rear wheels is formed of an injection molded plastic having a plurality of internal hollow portions and a metal band positioned to cover the hollow portions and form a support for an inner tube, wherein the injection molded plastic of the at least one rear wheel is formed by a process that involves removing one or more removable core elements.

(6) The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a perspective view of a shower/commode wheeled chair in accordance with some embodiments of this disclosure;

(2) FIGS. **2**A-**2**D are perspective views of a footrest and/or a footrest coupling that may be used in conjunction with a shower/commode wheeled chair according to some embodiments of this disclosure;

(3) FIGS. **2**E and **2**F are front perspective views of a front cross member comprising right and left portions and a clamp for supporting a footrest according to some embodiments of this disclosure;

(4) FIGS. **3**A-**3**C are side views of an armrest and/or an armrest support that may be used in conjunction with a shower/commode wheeled chair according to some embodiments of this disclosure;
(5) FIG. **3**D is a perspective views of portions of an armrest and an armrest support that may be used in conjunction with a shower/commode wheeled chair according to some embodiments of this disclosure;
(6) FIG. **3**E is a side perspective view of an armrest configured for coupling to a back support of a shower/commode wheeled chair according to some embodiments of this disclosure;
(7) FIG. **3**F is an exploded perspective view of a coupling for the armrest embodiment of FIG. **3**E;
(8) FIGS. **4**A and **4**B are perspective images of exemplary embodiments of a side guard that may be used in conjunction with a shower/commode wheeled chair according to some embodiments of this disclosure;
(9) FIGS. **5**A-**5**D are perspective views and images of a shower/commode wheeled chair and an adjustable lateral pad that may be used in conjunction with a shower/commode wheeled chair according to some embodiments of this disclosure;
(10) FIG. **5**E is a top perspective view of a lateral support configured for coupling to a back support of a shower/commode wheeled chair according to some embodiments of this disclosure;
(11) FIG. **5**F is an exploded perspective view of a link arrangement for use with the lateral support of FIG. **5**E;
(12) FIGS. **6**A-**6**C are perspective views of a shower/commode wheeled chair and a mechanism for adjusting a center of gravity of a shower/commode wheeled chair according to some embodiments of this disclosure;
(13) FIGS. **7**A and **7**B are perspective views of a shower/commode wheeled chair and a back support configured to fold into a generally horizontal arrangement to facilitate storing a shower/commode wheeled chair according to some embodiments of this disclosure;
(14) FIG. **7**C is a side perspective view of an adjustable/foldable back support for a shower/commode wheeled chair according to some embodiments of this disclosure;
(15) FIG. **7**D is an enlarged partially exploded view of an adjustable coupling for use with the adjustable/foldable back support embodiment of FIG. **7**C;
(16) FIGS. **8**A and **8**B are upper and lower perspective views, respectively, of a shower/commode seat for a shower/commode wheeled chair according to some embodiments of this disclosure;
(17) FIGS. **9**A-**9**C are a perspective view, a partial cross-sectional side view, and an exploded perspective view, respectively, of a rear wheel of a shower/commode wheeled chair according to some embodiments of this disclosure;
(18) FIG. **10**A is a perspective view of a back support for a shower/commode wheeled chair having folding handgrips according to some embodiments of this disclosure; and
(19) FIGS. **10**B-**10**D are enlarged side perspective views showing details of components of the folding handgrips shown in FIG. **10**A.

DETAILED DESCRIPTION

(20) This disclosure relates generally to shower/commode wheeled chairs that may be used by patients and elderly and/or injured individuals, and by caregivers for such individuals, in conjunction with movement or positioning relative to a shower or commode/toilet, for example for use during certain routine daily activities of life.
(21) FIG. **1** is a perspective view of a shower/commode wheeled chair **10** adapted to be locatable over a commode, but which could also be used in a shower setting, for example. The shower/commode wheeled chair **10** of FIG. **1** comprises a frame **12**, which may comprise a seat support portion **14** and a front portion **16** that extends downwardly from the seat support portion **14**. Frame **12**, including the seat support portion **14** and the front portion **16**, may be formed of a tubular material, for example. The tubular portions of frame **12** may, for example, be formed of a lightweight metal, such as aluminum, or of other metals, or of plastic or composite materials, for

example. The tubular nature of frame **12** may include hollow portions having circular, or semi-circular, or oval cross-sectional shapes, for example. An oval cross-sectional shape may, for example, provide advantages in terms of structural or weight-bearing support capabilities (e.g., an oval having its longer width directed vertically may provide the ability to support more weight than if oriented horizontally, for example). An oval shape may also provide advantages over a circular shape in areas where there is coupling between two components; for example, an oval shape would resist relative rotation between two radially coupled components as compared to a circular coupling.

(22) With continued reference to the shower/commode wheeled chair **10** of FIG. **1**, front portion **16** may comprise left and right frame elements **18**. Left and right frame elements **18** may extend downwardly and may present a generally vertical orientation, as is seen in FIG. **1**. In some embodiments, a front cross member **20** may be employed to couple the left and right frame elements **18** of the front portion **16** near a bottom portion of front portion **16**, according to some embodiments. Seat support portion **14** may comprise left and right frame elements **22** with generally horizontally oriented portions, as shown. In some embodiments, seat support portion **14** may also include a rear cross member **24** (see FIG. **6**B), which may be arranged to couple the left and right frame elements **22** of the seat support portion **14** near a rear portion thereof. The seat support portion **14** of the frame **12** may be supported (e.g., by a plurality of wheels **34**, **36** coupled to the shower/commode wheeled chair **10**) to thereby define a space beneath the seat support portion **14** to facilitate moving the shower/commode wheeled chair **10** over a commode/toilet, according to some embodiments.

(23) Back support **26** may be coupled to the seat support portion **14** in some embodiments and may comprise left and right frame elements **28** extending upwardly (e.g., generally vertically from seat support portion **14**), and may include a back support surface **30** extending between the left and right frame elements **28** of the back support **26**, for example. A shower/commode seat **32** may be disposed on and/or supported by the seat support portion **14** of the frame **12**, as shown in FIG. **1**. Shower/commode wheeled chair **10** may further comprise a plurality of wheels **34**, **36** coupled to the frame **12** and configured to movably support the frame **12** for transporting a patient seated thereon. In the embodiment shown in FIG. **1**, there are two front wheels **34** and two rear wheels **36**, although this is exemplary and other alternate configurations are contemplated by this disclosure.

(24) In some embodiments, shower/commode wheeled chair **10** may be configured to include one or more footrests **38** coupled to (including being releasably coupled to) the front portion **16** of a shower/commode wheeled chair **10** of this disclosure. Embodiments of a footrest **38** (and/or a footrest coupling **46**) for use with the shower/commode wheeled chair **10** are shown in the perspective views of FIGS. **2**A-**2**D. For example, FIG. **2**A shows a pair of footrests **38** (e.g., a left footrest **38** and a right footrest **38**) coupled to a front cross member **20** of shower/commode wheeled chair **10**. (The front cross member **20** is shown in isolation from the shower/commode wheeled chair **10** in FIGS. **2**A-**2**D.) In the particular embodiment shown in FIG. **2**A, each of the footrests **38** is coupled to an upwardly extending portion **40** of front cross member **20** (e.g., to a left and a right upwardly extending portion **40**). The footrest **38** may be pivotably coupled to the upwardly extending portion **40** to enable pivoting of the footrest **38** about a generally horizontal axis **42**, as seen in FIGS. **2**A and **2**B, for example. In some embodiments, footrest **38** may also be (or may alternatively be) rotatably coupled to the upwardly extending portion **40** to thereby enable rotation of footrest **38** about a generally vertical axis **44**, according to the embodiments shown in FIGS. **2**A-**2**C, for example.

(25) In the embodiments shown in FIGS. **2**A-**2**C, each of the footrests **38** are coupled to an upwardly extending portion **40** of a front cross member **20**. As shown, footrest **38** is configured to rotate about the respective upwardly extending portion **40** of the front cross member **20**; that is, footrest **38** rotates about a generally vertical axis **44** that is close to (e.g., coincident with) the centerline or axis of the associated upwardly extending portion **40**. This may, for example, result in

a more compact or convenient or efficient design as compared to other configurations where the footrest **38** is coupled at a location somewhat offset from the upwardly extending portion **40**.
(26) It should be noted that one or more footrests **38** could alternatively be coupled to a left frame element **18** or a right frame element **18** of the downwardly extending portion of front portion **16**. This alternate embodiment would allow footrest **38** to rotate about an axis generally formed by the right or left frame element **18** of the downwardly extending front portion **16**, and it would provide similar advantages as those described with respect to the embodiments depicted in FIGS. **2**A-**2**D above.
(27) With continued reference to FIGS. **2**A-**2**D, the footrest **38** may be coupled to an upwardly extending portion **40** of a front cross member **20** (or, in alternate embodiments, to the downwardly extending portion of front portion **16**) via a footrest coupling **46**, substantially as shown. For example, as shown in FIGS. **2**B-**2**D, footrest coupling **46** may comprise an upper portion **50** and a lower portion **52**. Upper portion **50** and lower portion **52** may each include an indexing feature **48** (see FIGS. **2**B and **2**C) that enables the footrest coupling **46** to be selectively rotated; that is, upper portion **50** may be selectively rotated relative to lower portion **52** as indicated by reference numeral **54** in FIG. **2**C (showing rotation of footrest **38** about axis **44** to move footrest **38** to an outward position to facilitate patient egress/ingress, for example). In some embodiments, the selective rotation of upper portion **50** relative to lower portion **52** may be effected or enabled by vertical displacement (e.g., vertical separation) between upper portion **50** and lower portion **52**; this may, for example, involve vertical displacement against spring tension that normally biases upper portion **50** toward lower portion **52** into a mating engagement, for example. Indexing feature **48**, as shown in FIGS. **2**B and **2**C may comprise a notched mating feature as shown, or some other type of shaped engagement that limits or prevents rotation of upper portion **50** and lower portion **52** relative to each other unless and until there is at least some level of vertical displacement therebetween. As shown, lower portion **52** remains securely fastened to upwardly extending portion **40** (e.g., it does not rotate from its original orientation) during rotation of upper portion **50** about the upwardly extending portion **40**. FIG. **2**D shows an additional or optional feature that may be part of footrest coupling **46**. For example, footrest coupling **46** may include a stop **56** shaped and positioned on footrest coupling **46** (e.g., formed in lower portion **52**) to facilitate pivoting and/or positioning (e.g., to limit downward movement when pivoting about horizontal axis **42**, for example) of footrest **38** into a configuration for supporting a patient's foot (as indicated by the downward pivoting of footrest **38** shown by reference numeral **58**). In some embodiments, footrest coupling **46** may be disposed coaxially with the right or left upwardly extending portions **40** (or, in alternate embodiments, disposed coaxially with a right or left frame element **18** of the downwardly extending portion of front portion **16**). In certain embodiments, the footrest coupling **46** may be configured to enable removal from frame **12** of the shower/commode wheeled chair **10**, for example, via a pushbutton release or other suitable release mechanism.
(28) FIGS. **2**E and **2**F are front perspective views of an embodiment of a wheeled chair **10** having a front cross member comprising right and left portions **20**A and **20**B, and a clamp **21** for coupling right and left cross member portions **20**A and **20**B together. In some embodiments, clamp **21** may also be used for supporting a footrest **38** according to some embodiments of this disclosure. FIG. **2**E illustrates an embodiment where the downwardly extending portion comprises a right frame element **18**A and a left frame element **18**B configured to receive the upwardly extending portions **40**A and **40**B of the front cross member to facilitate adjustment of the height of the lower portions of the cross members **20**A and **20**B and/or of a footrest **38** disposed thereon. As shown in the particular embodiment of FIG. **2**E, frame elements **18**A and **18**B and upwardly extending portions **40**A and **40**B taper inward as they extend toward a lower portion of each cross member **20**A, **20**B. When adjusting the height of the lower portion of cross members **20**A and **20**B and/or footrest **38** by varying the relative position of the frame elements **18**A and **18**B and upwardly extending portions **40**A and **40**B, the inward taper may cause an issue due to the varying width as you move

up and down, for example. The embodiment of FIG. **2**E is configured to address the varying width by providing a variable sized gap, "G," disposed between the innermost portions of right and left cross members **20**A and **20**B. For example, clamp **21** may have a number of fasteners (e.g., screws or bolts, not shown) disposed through holes (generally as shown in FIG. **2**F) and configured to be tightened to thereby tighten clamp **21** around the ends of cross members **20**A and **20**B to form a relatively stable platform, e.g., for supporting one or more footrests. The clamp **21** may likewise be loosened via the same fasteners to vary the size of the gap "G" in conjunction with adjusting the height of the footrest (e.g., by varying the up/down positioning of the upwardly extending portions **40**A and **40**B relative to the frame elements **18**A and **18**B).

(29) FIGS. **3**A-**3**D illustrate an embodiment of an arm rest **60** that may be used with the shower/commode wheeled chair **10** of FIG. **1**, for example. FIG. **3**A shows an arm rest **60** comprising an attachment portion **62** and an arm support portion **64** pivotably coupled to the attachment portion **62**. The arm rest **60** may be coupled to the frame **12** directly, or in some embodiments, an arm rest support **76** may be used to couple the arm rest **60** to the frame **12**. (Such an embodiment may enable adjustable positioning of the arm rest support **76** relative to the frame **12**, for example.) The attachment portion **62** may, for example, comprise a catch/latch mechanism **66** (e.g., incorporated in arm rest support **76**, or in a portion of frame **12**) which may be configured to enable releasable coupling and/or height adjustment of the arm rest **60** relative to the frame **12**. For example, catch/latch mechanism **66** may comprise a spring tensioned engagement mechanism configured to releasably engage a detent position **68** (e.g., one of the notches **68** formed in attachment portion **62** of the arm rest **60**). This arrangement may facilitate rapid up/down or vertical adjustment of the arm rest **60**, for example. Attachment portion **62** may include at least two detent positions **68** corresponding to at least two preset height adjustments for vertically adjusting the at least one arm rest **60**. In some alternate embodiments, the height/vertical adjustment of the one or more arm rests **60** may be configured to be continuously adjustable rather than employing discrete preset height adjustment levels in order to provide more flexible positioning options for patient comfort.

(30) The arm support portion **64** may comprise a spring release mechanism **70**, such as shown in the example of FIG. **3**B (in partial cross-section, corresponding to "inset **3**B" indicated in FIG. **3**A). For example, spring release mechanism **70** may be configured to selectively enable pivotable adjustment of the arm support portion **64** relative to the attachment portion **62** upon horizontal movement or displacement **73** of the arm support portion **64** by a certain amount **73**. For example, an internal catch or edge may prevent pivotable motion **74** (see FIG. **3**C) of arm support portion **64** unless and until arm support portion **64** has been moved sufficiently forward to position the catch beyond an internal obstruction, for example. In some embodiments, a small amount of forward-directed force **72** may be applied to arm support portion **64** to move it a required amount **73** to enable pivotable motion **74**, as illustrated in FIGS. **3**B and **3**C. Arm support portion **64** may be configured to pivot between a generally horizontal position and a generally vertical position as shown in FIG. **3**C when pivoted **74** following horizontal displacement **73**, for example.

(31) FIG. **3**D is a perspective view of a portion of arm rest **60** (e.g., with arm support portion **64** removed) that shows an attachment coupling **67** that may be employed according to some embodiments, for example, to hold catch/latch mechanism **66** into functional engagement with attachment portion **62** and/or any detent positions **68** formed therein. FIG. **3**D also shows an embodiment of arm rest support **76** that may be suited for coupling arm rest **60** to frame **12**. The arm rest support **76** of the exemplary embodiment of FIG. **3**D has a channel **78** formed in a lower portion thereof for coupling arm rest support **76** to frame **12** of wheeled chair **10**. In the particular example depicted in FIG. **3**D, the channel **78** may additionally be formed to engage with the contour of a frame **12** having an oval cross-sectional shape, for example, which may thereby improve stability and/or enhance structural integrity. The arm rest support **76** of the exemplary embodiment of FIG. **3**D may be a one-piece clamp having an oval-shaped (or partially oval-

shaped) channel **78** to help resist rotational movement of arm rest support **76** about a frame having a similar oval-shape. (A similar oval-shaped clamp and frame arrangement may be employed in other portions of wheeled chair **10**, such as in conjunction with attaching the footrest and/or footrest coupling **38**, **46** to the front portion of the frame.) Of course, embodiments of this disclosure may include one or more arm rests **60** coupled to shower/commode wheeled chair **10**, for example, a left arm rest **60** and a right arm rest **60**.

(32) In some embodiments, arm rest **60** may alternatively be operably coupled to back support **26**. For example, attachment portion **62** of arm rest **60** may be releasably coupled and/or adjustably coupled to right and left frame elements **28** of back support **26**, according to some embodiments. The coupling may comprise a locking pin or comparable mechanism that may enable removal of the arm rest **60** from back support **26** and/or may enable angular adjustment of arm rest **60** relative to frame elements **28** of back support **26**.

(33) FIG. **3**E is a side perspective view of an embodiment of an arm rest **364** configured for coupling to a frame element **328** of a back support of a shower/commode wheeled chair, according to some embodiments of this disclosure. A number of benefits may be derived from the embodiment shown in FIG. **3**E. For example, the overall width of the wheeled chair **10** may be reduced or minimized by employing one or more of the features illustrated in FIG. **3**E. For example, frame element **328** may comprise a lower slot **329** for receiving an upper portion of support **376**, thereby reducing the width of frame element **328** that extends outwardly beyond support **376** in such an embodiment. (Support **376** may be attached to, or integrally formed with, frame element **322** of the seat support portion of wheeled chair **10**, according to various embodiments.) Also, arm rest **364** may be coupled to frame element **328** of the back support, rather than to support **376**, which may also reduce the overall width. For example, as shown, arm rest **364** may be coupled to frame element **328** via arm rest coupling **366**. Arm rest coupling **366** is configured to enable adjustment of the height, “H,” of arm rest **364** along frame element **328** in a continuous range of heights, for example. Arm rest coupling **366** may also be configured to be removed, or releasably pivoted upward (e.g., near vertical), or pivoted into series of defined angles, “A,” that may correspond to a series of defined angles of the back support (e.g., to retain a mostly horizontal position of arm rest **364** to complement a number of back angle positions).

(34) FIG. **3**F is an exploded partial perspective view of the arm rest coupling **366** used with the arm rest **364** embodiment depicted in FIG. **3**E. As shown, arm rest coupling **366** may comprise a clamp portion **368** configured to be tightened and loosened around a frame element **328** (indicated by dashed lines) of the back support to enable height adjustment of the arm rest **364** over a continuous range of heights, “H,” along frame element **328**. Arm rest coupling **366** may also comprise an arm rest pivot portion **367**, pivotably coupled to clamp portion **368**, and operatively coupled to arm rest **364** so that arm rest **364** can be pivoted relative to clamp portion **368**. In some embodiments, clamp portion **368** may include a release lever **369** configured to, upon actuation, enable complete removal of arm rest **364** from wheeled chair **10**, along with removal of arm rest pivot portion **367** from clamp portion **368** in some embodiments. Release lever **369** may also enable pivotable movement of arm rest **364** over a series of angular adjustment positions via movement of a pin **371** slidably received within a slot **370** formed in arm rest pivot portion **367**. Finally, it should be noted that such an embodiment may also retain the “push forward, tilt upward” feature described above with respect to FIGS. **3**A-**3**C.

(35) FIGS. **4**A and **4**B are perspective images showing embodiments of a side guard **80** that may be used with a shower/commode wheeled chair **10** according to some embodiments of this disclosure. Side guard **80** may be removably coupled to the frame **12** adjacent to the seat support portion **14** of the frame **12**. In some embodiments, side guard **80** may be configured to be adjusted vertically relative to seat support portion **14** or to other portions of frame **12**, for example, via a slidable friction fit. Partially shown in FIG. **4**A is a side guard mounting post **82**, which may be used to effect a slidable friction fit with a corresponding portion of the frame **12**, or with an attachment or

coupling thereto. In some embodiments, one or more adjustment slots **84** may be formed in side guard **80** (as best seen in FIG. **4**B) to facilitate releasable coupling of side guard **80** to mounting post **82**. A particular adjustment slot **84** may enable some amount of vertical adjustment in the positioning of side guard **80**, for example. Similarly, a plurality of adjustment slots **84** positioned side by side (as shown in FIG. **4**B) may facilitate some amount of forward and backward adjustment of the side guard **80** relative to the seat support portion **14** or other portions of frame **12**, according to some embodiments.
(36) FIGS. **5**A-**5**D show perspective views and an image of a shower/commode wheeled chair **10** that may be configured to further include at least one lateral pad **86** for providing lateral support to a patient seated on chair **10**, for example. Such lateral support may be desirable to keep a patient from leaning and/or falling sideways from a seated position on wheeled chair **10**. In such embodiments, lateral pad **86** may form a cushioned surface with a generally vertical orientation configured to be placed against a side of a patient (e.g., near the right and left sides of their chest/back area). FIG. **5**A shows a back support **26** of chair **10** with potential locations (e.g., on left and right frame elements **28** of back support **26**, e.g., near back support surface **30**) for coupling a lateral pad **86** thereto. In some embodiments, there may be one or more lateral pads **86** coupled to back support **26**, such as that shown in FIG. **5**B. In some embodiments, lateral pad **86** may have a release mechanism **88** (e.g., a push-button activated release **88**, as seen in FIGS. **5**C and **5**D) that, when activated, enables pivotably positioning the lateral pad **86** relative to the back support **26** from a patient contact/support position to an outward position **90**, as shown in FIG. **5**D to facilitate patient egress and ingress, for example. FIGS. **5**C and **5**D also show adjustable fasteners **96** and **98**, respectively. Adjustable fastener **96**, for example, may be used to adjust lateral pad **86** in a forward/backward direction **92** (relative to wheeled chair **10**), as shown in FIG. **5**C. Adjustable fastener **98**, for example, may be used to couple the lateral pad **86** to a portion of back support **26** (e.g., to left or right frame element **28**), with the ability to adjust the positioning of lateral pad **86** in both an upward/downward direction (e.g., by varying placement vertically along a left or right frame element **28** of back support **26**) or in a lateral (e.g., inward/outward) direction **94**, as shown in FIG. **5**D, for example. In some embodiments, additional links (or arms) may be added to one or more of the lateral pads **86**, for example, pivotably coupled via additional release buttons **88**. This may be helpful to accommodate a wider range of patient sizes, according to some embodiments.
(37) FIG. **5**E is a top perspective view of another embodiment of a lateral pad **386** configured for coupling to a back support of a shower/commode wheeled chair according to some embodiments of this disclosure. In the embodiment depicted, lateral pad **386** is coupled to the right and/or left frame element **328** (partially shown in dotted lines to allow more details of the support arrangement to be shown) via an adjustable fastener **398**, configured to adjust the position of lateral pad **386** in an upward/downward direction by loosening and tightening adjustable fastener **398** about frame element **328**. Also shown in FIG. **5**E is a pushbutton release **388**, which can operate against spring tension (not shown) to enable pivotable positioning of the lateral pad **386** to an outward position (indicated by arrow **390**) relative to the back support portion of the wheeled chair. Pushbutton **388** may be housed within a portion of adjustable fastener **398** and can be further configured to support and control pivotable actuation of a lateral pivot link **392** in the direction generally indicated by arrow **390**, for example. In some embodiments, actuation of pushbutton release **388** quickly enables pivotably repositioning lateral pad **386** to swing out and away from a patient to facilitate rapid egress and ingress of a patient. In some embodiments, the remaining links forming an “arm” of the lateral support arrangement may be adjusted and secured for a particular patient's size and comfort needs so that, when lateral pad **386** is swung back from an outward egress position into a patient support position, it can quickly be returned to its custom, adjusted, patient-specific position.
(38) In the embodiment of FIG. **5**E, a series of links forming the arm of the lateral support arrangement may include the aforementioned lateral pivot link **392**, a dovetail link **394** coupled to the lateral pivot link **392** (e.g., via a tongue/groove or comparable friction fit arrangement), and one

or more lateral links **396** coupled distal to the dovetail link **394**. FIG. **5**F is an exploded perspective view of exemplary links that may form portions of the lateral support arrangement of FIG. **5**E. For example, in FIG. **5**F, the angular positioning of lateral link **396** relative to dovetail link **394** can be varied and secured in place via a fastener (e.g., screw, bolt/nut, etc., not shown) engaged through corresponding openings of links **394** and **396** aligned along a generally vertical axis **399**, as shown. In some embodiments, the angular adjustment of links **394** and **396** may be enhanced by the use of corresponding mating surfaces **395** and **397** (e.g., shaped, radial ridges) of links **396** and **394**, respectively, that can align and thereby provide a more secure engagement of the various links according to a patient-specific adjustment of relative angles of the links. As noted above, additional lateral links **396** may be added in series as needed to accommodate patients of varying sizes.

(39) FIGS. **6**A-**6**C are perspective views of a shower/commode wheeled chair **10** having a center of gravity adjustment mechanism **100** (FIGS. **6**B and **6**C) that may be configured to move the positioning of the rear wheels **36** of shower/commode wheeled chair **10** relative to other elements of wheeled chair **10**. For example, the shower/commode wheeled chair **10** depicted in FIG. **6**A may comprise at least one center of gravity adjustment mechanism **100** (e.g., a left and a right center of gravity adjustment **100**), wherein the center of gravity adjustment **100** is configured to move one or more of the at least two rear wheels **36** forward/backward and/or upward/downward relative to the seat support portion **14**. FIG. **6**B is a partial exploded perspective view of a shower/commode wheeled chair **10** having a seat support portion **14** formed of left and right frame elements **22** coupled together with a rear cross member **24** near a rear portion of the seat support portion **14**. FIG. **6**C is an enlarged partial view of the inset from FIG. **6**B showing details of a center of gravity adjustment **100** coupled to a frame element **22** of seat support portion **14**. In the particular embodiment depicted, center of gravity adjustment **100** is coupled within a slot formed in frame element **22**. In some embodiments, frame element **22** may have two slots formed therein to couple and/or adjust positioning of the center of gravity adjustment mechanism **100**. For example, center of gravity adjustment **100** may extend downward from a lower slot (not shown) and may be coupled or fastened to frame element **22** via one or more fasteners disposed adjacent a side-facing slot formed in frame element **22** to facilitate forward and/or backward adjustment of center of gravity adjustment **100** (as indicated by “FIR” in FIG. **6**C). Additionally, or alternately, center of gravity adjustment **100** may itself have a generally vertically-oriented slot formed therein to facilitate adjustment of the height of the coupling of rear wheels **36** (e.g., via one or more fasteners, such as bolts, etc.) to the center of gravity adjustment **100** (as indicated by “LA)” in FIG. **6**C).

(40) FIGS. **7**A and **7**B are perspective views of a shower/commode wheeled chair **10** having a back support **26** that is pivotably coupled to the seat support portion **14** to enable folding the back support **26** into a compact configuration that may be suited for storage, for example. FIG. **7**A indicates a folding direction **102** that may be actuated by a user (e.g., a patient and/or a caregiver) to place the back support **26** into a position generally parallel to the seat support portion **14**, for example, upon actuation of one or more release actuators **104** (see FIG. **7**B). For example, as shown in the exploded partial view of FIG. **7**B, a release actuator **104** may comprise a pushbutton release that enables a left and/or right frame element **28** to pivot downward/forward (as indicated by reference numeral **102**). In some embodiments, release actuator **104** may facilitate adjusting the angle of back support **26**, for example to improve patient comfort. In such an embodiment, release actuator **104** may enable adjusting the angle of back support **26** via pivotable motion of frame elements **28** into a number of predefined detent positions, according to some embodiments. Left or right frame element **28** is shown pivotably coupled to arm rest support **76** in the particular example of FIG. **7**B, but it could be alternatively coupled to left or right frame element **22**, or to rear cross member **24**, etc. In some embodiments, one or more release actuators **104** may be further configured to enable vertical adjustment of back support **26**. For example, a pushbutton release may enable left and/or right frame elements **28** of back support **26** to move slidably upward or downward relative to seat support portion **14** upon actuation of the pushbutton release, or to fully

release the frame elements **28** from the seat support portion **14**. In some cases, the pushbutton release may be the same release actuator **104** that enables pivotable folding of the back support **26** via pivotable coupling at arm rest support **76**, or it may be a separate release mechanism. In certain embodiments, it may be desirable to enable removal of an arm rest **60** to facilitate folding back support **26** downward into a storage position; in such embodiments, a single pushbutton release may be employed to facilitate rapid release and removal of the arm rest **60** prior to folding of the back support **26**.

(41) FIG. **7**C is a side perspective view of another embodiment of an adjustable/foldable back support for a shower/commode wheeled chair. (FIG. **7**C happens to show the elements corresponding to the left side of wheeled chair **10**, with it being understood that the right side of wheeled chair **10** would have equivalent corresponding elements for this embodiment.) Frame element **328** of the back support is shown being configured to pivotably move to vary the angle of the back support over a range of angles indicated by curved arrow **302** in FIG. **7**C. In the embodiment shown, frame element **328** is pivotably supported near its lower end by support **376**. In some embodiments, support **376** may be coupled to, or integrally formed with, frame element **322** of the seat support portion of wheeled chair **10**. A back angle release lever **304** is also shown in FIG. **7**C. Actuation of back angle release lever **304** may enable repositioning or adjusting the angle of back support (e.g., by changing the angle of the frame elements **328** of the back support). In some embodiments, release lever **304** may enable adjustment of the back angle to a number of predefined positions for patient comfort (e.g., slight adjustments forward and back from a mostly vertical position). Alternatively, or additionally, release lever **304** may enable pivoting or folding the back support of wheeled chair **10** all the way down into a mostly horizontal position (e.g., generally parallel to the seat support portion of wheeled chair **10**).

(42) FIG. **7**D is an enlarged partially exploded view of an adjustable coupling for use with the adjustable/foldable back support embodiment of FIG. **7**C. For example, back angle extension **305** may be used to couple frame element **328** to release lever **304**, according to some embodiments, although release lever **304** could optionally be coupled to frame element **328** without using extension **305**. Release lever **304** is pivotably coupled to extension **305** in the particular embodiment shown and may be spring-tensioned to facilitate releasable engagement of a tab **306** with one or more notches or detent positions **378** associated with support **376**. In the embodiment depicted in FIG. **7**D, the detent positions **378** may correspond to a series of predefined back angles, such as vertical (0 degrees), slight recline (−7 degrees back), and recline (−14 degrees back), as possible examples. The predefined back angles could vary, of course, but would typically fall in a range within about 45 degrees from vertical in most cases. An additional notch or position **378** may allow folding or pivoting the back support fully forward into a mostly horizontal orientation, which may be suitable for storage of wheeled chair, according to some embodiments.

(43) FIGS. **8**A and **8**B are upper and lower perspective views, respectively, of a shower/commode seat **32** that may be used in conjunction with a shower/commode wheeled chair **10** of this disclosure. FIGS. **8**A and **8**B together show a shower/commode seat **32** comprising an upper padded side **108**, a lower base side **110** (see FIG. **8**B), and a cover material **106** that extends over the upper padded side **108** and is secured to the lower base side **110**. In some embodiments, cover material **106** may be secured using a plurality of fasteners positioned within a recessed groove **112** formed in the lower base side **110**. For example, the plurality of fasteners may include a number of staples, brads, nails, screws, etc. (not shown in FIGS. **8**A and **8**B), that are applied through the cover material **106** and lodged into the recessed groove **112** of the lower base side **110** to secure the cover material **106** to the shower/commode seat **32**. In some embodiments, a sealant is applied over the plurality of fasteners after placement/positioning of the fasteners within the recessed groove **112**.

(44) FIGS. **9**A-**9**C illustrate a rear wheel **36** that may be used with a shower/commode wheeled chair **10** according to some embodiments of this disclosure. As noted above, a shower/commode

wheeled chair **10** may include a plurality of wheels, including at least two rear wheels **36** coupled to the frame **12** (e.g., directly and/or indirectly, including via the use of a center of gravity adjustment **100** as described hereinabove), and configured to movably support the frame **12**. In some embodiments, one or more of the rear wheels **36** may comprise an injection molded material **114** (e.g., a plastic and/or composite material), as shown in FIG. **9**A. The injection molded material **114** of rear wheel **36** may form a plurality of internal hollow portions **116** as a result of the injection molding process, as shown. For example, as shown in FIG. **9**B, a number of removable core elements **120**, may be present during the injection molding process, and may be removed upon completion of injection molding to leave behind the plurality of hollow portions **116**. In certain embodiments, the number of removable core elements **120** may range from two to six or more, depending on the nature of the wheel **36** (e.g., number of spokes formed, etc.). The hollow portions **116** may contribute to reducing the overall weight of wheel **36**, while retaining the structural strength and/or integrity of the material used in the forming process. A band **118** (e.g., a thin metal band **118**, for example), as shown in FIG. **9**C, may then be disposed to cover the plurality of hollow portions **116** of the molded wheel **36**. The use of band **118** may form a support surface for placement of an inner tube **122** thereabout. The use of band **118** may also serve to prevent inner tube **122** from being pressed into the plurality of hollow portions **116**, which may damage inner tube **122** and/or reduce the effectiveness of inner tube **122** at providing a smooth or comfortable ride to a patient seated thereon, for example.

(45) FIG. **10**A is a perspective view of a back support **26** for a shower/commode wheeled chair **10** having folding handgrips **202** according to some embodiments of this disclosure. Handgrips **202** may facilitate moving, pushing/pulling, and maneuvering wheeled chair **10** by a user or assistant. Handgrips **202** may be disposed at or near an upper portion of back support **26**. For example, a left handgrip **202** and a right handgrip **202** may be disposed at an upper portion of respective left and right frame members **28** of back support **26**, as shown in FIG. **10**A. In some embodiments, handgrips **202** may be folding or foldable handgrips **202** configured to fold downward or inward; arrow **206** in FIG. **10**A shows a possible folding path for a foldable handgrip **202** that can fold downward, for example, pivoting about a generally horizontal axis. Alternatively, foldable handgrips **202** could be configured to fold inward, or upward, or in some other direction. The folding of handgrips **202** may be beneficial, for example, in positioning the shower/commode wheeled chair directly over a commode/toilet for patient use. The folding of handgrips **202** can thereby lessen the rearward protrusion (or the horizontal footprint) of the shower/commode wheeled chair **10** to enable moving it more fully or directly over a commode in some circumstances. In the particular embodiment depicted in FIG. **10**A, a button **204** is disposed on the underside of folding handgrip **202** to unlock the handgrip and enable folding; positioning of the button **204** could be on the underside of folding handgrip **202**, as shown in the embodiment of FIG. **10**A, or could be disposed in other locations or orientations, although placing button **204** underneath the handgrip **202** may be helpful to avoid ingress of debris, dirt, or liquids into the button mechanism, which may be undesirable.

(46) FIGS. **10**B-**10**D are enlarged side perspective views showing details of various components of the folding handgrips **202** of FIG. **10**A. FIG. **10**B is an enlarged image showing the relative positioning of a handgrip **202** both before folding and after folding downward along folding path **206**. The folding is enabled by a pushbutton release (e.g., by pushing button **204**) to release a catch mechanism that enables folding of the handgrip **202**. When returned to the original horizontal orientation, the handgrip **202** engages a catch and “clicks” back into its normal position. FIG. **10**C is a partially exploded view showing certain aspects of the pushbutton release mechanism **204**. As shown, upward motion of pushbutton **204** when pressed causes one or more actuating rods **210** to be moved generally horizontally by action of an angled slot **212** formed in button **204**; angled slot **212** slidably receives the one or more actuating rods **210**. When the actuating rods **210** are moved by actuation of the pushbutton **204**, one of the actuating rods **210** disengages from a notch (e.g., a

depression, indentation, slot, etc.) formed in a portion of handgrip **202**, thereby enabling folding (e.g., pivoting of the handgrip **202**, downward in the example illustrated). FIG. **10**D is an enlarged top perspective view of an exemplary actuation member **211**, having a pair of actuating rods **210** extending laterally therefrom, and configured to be actuated upon pressing of button **204** upward. Other comparable mechanisms for effectuating a releasable pivotable (folding) handgrip arrangement are contemplated and would be deemed to be within the scope of this disclosure.
(47) Various examples have been described with reference to the accompanying drawing figures. These and other alternatives and variations are contemplated and are deemed to be within the scope of the following claims.

Claims

1. A shower/commode wheeled chair adapted to be locatable over a commode comprising: a frame comprising a seat support portion and a front portion extending downwardly from the seat support portion, the front portion comprising left and right frame elements and a front cross member coupling the left and right frame elements of the front portion near a bottom portion thereof, the seat support portion comprising left and right frame elements and a rear cross member coupling the left and right frame elements of the seat support portion near a rear portion thereof; a back support coupled to the seat support portion, the back support comprising left and right frame elements and a back support surface extending between the left and right frame elements of the back support; a shower/commode seat supported by the seat support portion of the frame; a plurality of wheels coupled to the frame and configured to movably support the frame and to define a space beneath the seat support portion to facilitate moving the shower/commode wheeled chair over a commode; and at least one footrest coupled to the front portion, the at least one footrest being pivotably coupled to pivot about a generally horizontal axis, the footrest further being rotatably coupled to rotate about a generally vertical axis.
2. The shower/commode wheeled chair of claim 1 wherein the at least one footrest comprises a left footrest and a right footrest.
3. The shower/commode wheeled chair of claim 1 wherein the at least one footrest is configured to be removable from the front portion of the frame.
4. The shower/commode wheeled chair of claim 1 wherein the at least one footrest is rotatably coupled to rotate about either the right frame element or the left frame element of the downwardly extending front portion.
5. The shower/commode wheeled chair of claim 1 wherein the at least one footrest is rotatably coupled to rotate about either a right upwardly extending portion or a left upwardly extending portion of the front cross member.
6. The shower/commode wheeled chair of claim 1 wherein the at least one footrest is coupled to the front portion via a footrest coupling, the footrest coupling comprising an upper portion and a lower portion, the upper and lower portions each including an indexing feature, wherein the indexing feature of the footrest coupling is configured to selectively enable rotation of the at least one footrest about the right or left upwardly extending portion of the front cross member via vertical displacement of the upper portion relative to the lower portion.
7. The shower/commode wheeled chair of claim 1 wherein the front cross member comprises a right front cross member, a left front cross member, and a clamp configured to clamp the right front cross member to the left front cross member with a gap therebetween.
8. The shower/commode wheeled chair of claim 7 wherein the right front cross member and the left front cross member each has an upwardly extending portion that tapers such that the gap between the right front cross member and the left front cross member varies in size when a height of the right and left front cross members is adjusted, and wherein the clamp is configured to clamp the right front cross member to the left front cross member to accommodate changes in gap size.

9. A shower/commode wheeled chair adapted to be locatable over a commode comprising: a frame comprising a seat support portion and a front portion extending downwardly from the seat support portion, the front portion comprising left and right frame elements and a front cross member coupling the left and right frame elements of the front portion near a bottom portion thereof, the seat support portion comprising left and right frame elements and a rear cross member coupling the left and right frame elements of the seat support portion near a rear portion thereof; a back support coupled to the seat support portion, the back support comprising left and right frame elements and a back support surface extending between the left and right frame elements of the back support; a shower/commode seat supported by the seat support portion of the frame; a plurality of wheels coupled to the frame and configured to movably support the frame and to define a space beneath the seat support portion to facilitate moving the shower/commode wheeled chair over a commode; and at least one arm rest releasably coupled to the wheeled chair, the arm rest comprising an attachment portion and an arm support portion pivotably coupled to the attachment portion, the attachment portion configured to adjust a height of the arm rest, the arm support portion comprising a spring release mechanism configured to selectively enable pivotable adjustment of the arm support portion relative to the attachment portion upon horizontal displacement of the arm support portion.
10. The shower/commode wheeled chair of claim 9 wherein the attachment portion comprises a catch/latch mechanism configured to operably couple the arm rest to the frame of the wheeled chair.
11. The shower/commode wheeled chair of claim 9 wherein the attachment portion comprises a clamp portion configured to releasably couple to one of the left and right frame elements of the back support at an adjustable height, and wherein the arm support portion comprises a pivot portion pivotably coupled to the clamp portion.
12. The shower/commode wheeled chair of claim 11 wherein the clamp portion comprises a release lever configured to, upon actuation, either: (a) enable removal of the arm rest from the wheeled chair, or (b) enable pivotable adjustment of the arm rest relative to one of the left and right frame elements of the back support.
13. The shower/commode wheeled chair of claim 11 wherein the left and right frame elements of the back support are pivotably coupled to the seat support portion to enable folding the back support into a horizontal configuration generally parallel to the seat support portion upon actuation of one or more release actuators.
14. The shower/commode wheeled chair of claim 11 wherein the left and right frame elements of the back support are pivotably coupled to the seat support portion to enable adjusting an angle of the back support upon actuation of one or more release actuators.
15. The shower/commode wheeled chair of claim 9 further comprising at least one lateral pad operably coupled to one of the left and right frame elements of the back support, the at least one lateral pad configured to provide lateral support to a patient seated on the wheeled chair.
16. The shower/commode wheeled chair of claim 15 wherein the at least one lateral pad is operably coupled to one of the left and right frame elements of the back support via a link arrangement, the link arrangement comprising: (a) a first link pivotably coupled to one of the left and right frame elements of the back support, the first link having a pushbutton release configured to enable pivotable rotation of the first link about one of the left and right frame elements of the back support upon actuation of the pushbutton release; and (b) at least one lateral link coupled to the first link at an adjustable angle therebetween, the adjustable angle between the lateral link and the first link being at least partially maintained by one or more corresponding mating surfaces formed between the lateral link and the first link.
17. The shower/commode wheeled chair of claim 9 further comprising one or more handgrips disposed near an upper portion of the wheeled chair.
18. The shower/commode wheeled chair of claim 17 wherein the one or more handgrips is

configured to fold about a pivot axis upon actuation of a release button.
19. A shower/commode wheeled chair adapted to be locatable over a commode comprising: a frame comprising a seat support portion and a front portion extending downwardly from the seat support portion, the front portion comprising left and right frame elements and a front cross member coupling the left and right frame elements of the front portion near a bottom portion thereof, the seat support portion comprising left and right frame elements and a rear cross member coupling the left and right frame elements of the seat support portion near a rear portion thereof; a back support coupled to the seat support portion, the back support comprising left and right frame elements and a back support surface extending between the left and right frame elements of the back support; a shower/commode seat supported by the seat support portion of the frame; and a plurality of wheels coupled to the frame and configured to movably support the frame and to define a space beneath the seat support portion to facilitate moving the shower/commode wheeled chair over a commode wherein the left and right frame elements of the back support are configured to be pivotably adjusted to a range of back angles via actuation of a back angle release lever, the range of back angles including: (a) a storage configuration with the left and right frame elements of the back support pivoted forward to a generally horizontal position; and (b) at least two patient preference back angles.
20. The shower/commode wheeled chair of claim 19 wherein the at least two patient preference back angles of the back support comprises three predefined back angles within 45 degrees of vertical.
21. The shower/commode wheeled chair of claim 19 further comprising an arm rest pivotably coupled to one of the left and right frame elements of the back support, the arm rest being coupled to the back support via an arm rest coupling, the arm rest coupling comprising a clamp portion configured to adjust a height of the arm rest relative to the back support, the arm rest coupling further comprising a pivot portion pivotably coupled to the clamp portion, the pivot portion configured to pivot relative to the clamp portion.
22. The shower/commode wheeled chair of claim 21 wherein the pivot portion of the arm rest coupling is configured to adjust an angle of the arm rest relative to the back angle of the back support to maintain a generally horizontal orientation of the arm rest.
23. The shower/commode wheeled chair of claim 21 wherein the arm rest is configured to be released from the clamp portion to facilitate pivotably adjusting the back support to the storage configuration.
24. The shower/commode wheeled chair of claim 21 wherein the arm rest further comprises an arm support portion having a spring release mechanism configured to selectively enable pivotable motion of the arm support portion relative to the clamp portion upon horizontal displacement of the arm support portion.
25. The shower/commode wheeled chair of claim 19 further comprising one or more handgrips disposed near an upper portion of the wheeled chair, the one or more handgrips being configured to fold about a pivot axis upon actuation of a release button.
26. The shower/commode wheeled chair of claim 19 further comprising at least one lateral pad operably coupled to one of the left and right frame elements of the back support, the at least one lateral pad configured to provide lateral support to a patient seated on the wheeled chair, the at least one lateral pad being operably coupled to one of the left and right frame elements of the back support via a link arrangement, the link arrangement comprising: (a) a first link pivotably coupled to one of the left and right frame elements of the back support, the first link having a pushbutton release configured to enable pivotable rotation of the first link about one of the left and right frame elements of the back support upon actuation of the pushbutton release; and (b) at least one lateral link coupled to the first link at an adjustable angle therebetween, the adjustable angle between the lateral link and the first link being at least partially maintained by one or more corresponding mating surfaces formed between the lateral link and the first link.

27. A shower/commode wheeled chair adapted to be locatable over a commode comprising: a frame comprising a seat support portion and a front portion extending downwardly from the seat support portion, the front portion comprising left and right frame elements and a front cross member coupling the left and right frame elements of the front portion near a bottom portion thereof, the seat support portion comprising left and right frame elements and a rear cross member coupling the left and right frame elements of the seat support portion near a rear portion thereof; a back support coupled to the seat support portion, the back support comprising left and right frame elements and a back support surface extending between the left and right frame elements of the back support; a shower/commode seat supported by the seat support portion of the frame; a plurality of wheels coupled to the frame and configured to movably support the frame; and at least one arm rest releasably coupled to the frame, the arm rest comprising an attachment portion and an arm support portion pivotably coupled to the attachment portion, the attachment portion comprising a catch/latch mechanism that enables releasable coupling and height adjustment of the arm rest relative to the frame, the arm support portion comprising a spring release mechanism configured to selectively enable pivotable adjustment of the arm support portion relative to the attachment portion upon horizontal displacement of the arm support portion.
28. The shower/commode wheeled chair of claim 27 wherein the catch/latch mechanism of the attachment portion includes at least two preset height adjustments for vertically adjusting the at least one arm rest.
29. The shower/commode wheeled chair of claim 27 wherein the arm support portion is configured to pivot between a generally horizontal position and a generally vertical position.
30. A shower/commode wheeled chair adapted to be locatable over a commode comprising: a frame comprising a seat support portion and a front portion extending downwardly from the seat support portion, the front portion comprising left and right frame elements and a front cross member coupling the left and right frame elements of the front portion near a bottom portion thereof, the seat support portion comprising left and right frame elements and a rear cross member coupling the left and right frame elements of the seat support portion near a rear portion thereof; a back support coupled to the seat support portion, the back support comprising left and right frame elements and a back support surface extending between the left and right frame elements of the back support; a shower/commode seat supported by the seat support portion of the frame; a plurality of wheels coupled to the frame and configured to movably support the frame; and at least one lateral pad coupled to the back support, the at least one lateral pad having a release mechanism for pivotably positioning the lateral pad relative to the back support.
31. The shower/commode wheeled chair of claim 30 wherein the release mechanism for the at least one lateral pad is a pushbutton release that enables pivoting the lateral pad about the pushbutton release to an outward position to facilitate patient egress, and wherein the lateral pad is further configured to be adjusted inward/outward and forward/backward.
32. A shower/commode wheeled chair adapted to be locatable over a commode comprising: a frame comprising a seat support portion and a front portion extending downwardly from the seat support portion, the seat support portion comprising left and right frame elements and a rear cross member coupling the left and right frame elements of the seat support portion near a rear portion thereof; a back support coupled to the seat support portion; a shower/commode seat supported by the seat support portion of the frame; and a plurality of wheels coupled to the frame and configured to movably support the frame, the plurality of wheels comprising at least two rear wheels, wherein at least one of the rear wheels comprises an injection molded plastic having a plurality of internal hollow portions, and a metal band disposed to cover the plurality of internal hollow portions and form a support for an inner tube, wherein the injection molded plastic of the at least one rear wheel is formed by a process including removing one or more removable core elements.